How to select K?

22 May 2023 19:42

$\boxed{K=?} \rightarrow K=3 \quad K=5$

↓

data

heuristic (jugaad)

experimentation

$\sqrt{n}$ $n \rightarrow$ observation

$\boxed{400} \rightarrow \sqrt{400} = 20$

Session on KNN Part 1 HOME INSERT DRAW VIEW

Stop

(jugaad)

$\sqrt{n}$ $n \to$ observation

$\boxed{400} \to \sqrt{400} = 20$

even value for k

$20 \to \underbrace{19 \to 21}$

cross-validation

$\underbrace{1000} = n$

800 (train) train

200 (test)

run $\to$ accuracy

diff knn

$\boxed{knn \to 1} \to 83$

$\boxed{knn \to 2} \to 81$

$knn \to 3 \to 87$

$\boxed{knn \to 25 \to 91}$

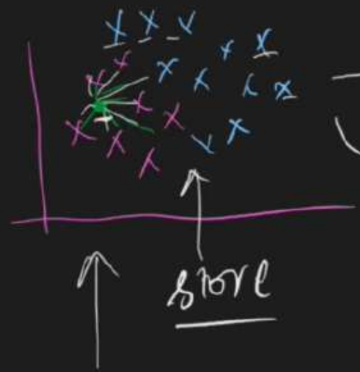
failure case

(500000, 100)

1) large datasets → $n = 5L$, $f = 100$

↳ knn is a lazy learning techn

prediction
↓
slow → dataset



→ query point → prediction

↳ training → nothing

5L distance → sort → majority →

store

low latency $\boxed{3\text{ sec}}$ → slows

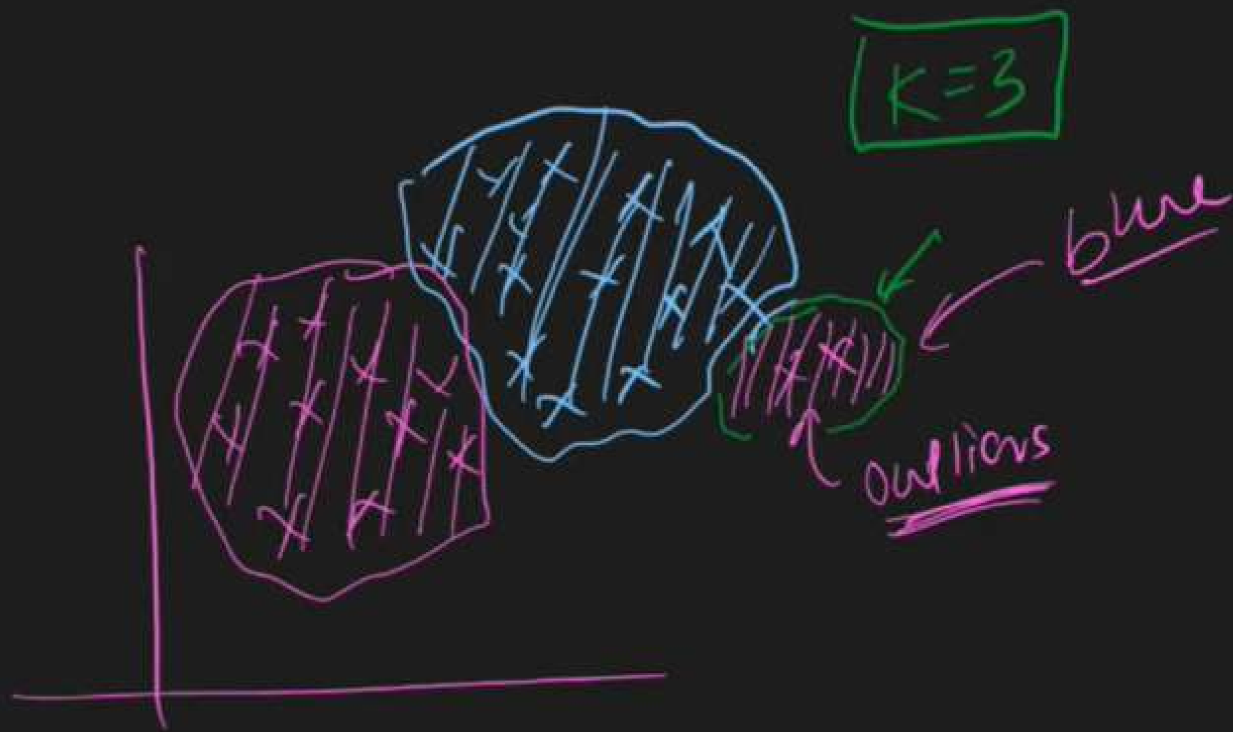
2) High dim data → $\boxed{f = 500}$ →

↓

curse of dimension → distance concept → reliable

knn

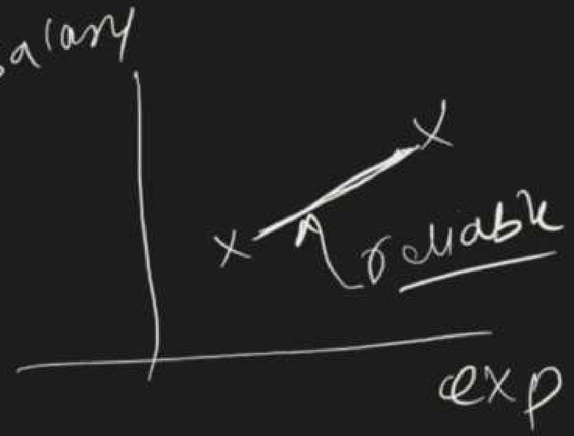
3> Outliers



4) Non- homogeneous scales

dominate

exp	salary	fire
0-25	20k-1cr	



→ scaling

5) Imbalanced dataset

↳ Yes → 98%

No → 2%

↳ biased → pink